INF10025 Week 8 – SQL and Modelling (Part 4)

Practice Questions · Lecture 8 Parts 1–3 · movie database and example tables

How to use this practice set

Open **INF10025_Week08_Practice.accdb** (same folder) and look at the tables in the Navigation Pane, or run the statements in Oracle iSQL*Plus Junior using your own table names (e.g. add your 5520 suffix: MOVIE5520, ACTOR5520). Write each answer yourself first, then compare with **INF10025_Week08_Practice_Answers.pdf**. Tags show the level: Pass, Credit, Distinction.

Oracle vs Access: the unit uses Oracle SQL. In Access, dates are written #1970-01-01#, TO_DATE/TO_CHAR do not exist (use Format()), INTERSECT and MINUS are not supported (use IN / NOT IN), and CHECK constraints cannot be created with the query designer.

Tables in the Access file

Table	Columns	Rows	Used for
MOVIE	MOVIENO (PK), TITLE, RELYEAR, RUNTIME, RATINGCODE (FK), COLOURCODE (FK), TMDB_SCORE, TMDB_VOTES, IMDB_ID	395	Movie database from the laboratory (copied as is)
ACTOR	ActorNo (PK), FullName, GivenName, Surname, Gender, Birthdate, BirthCountry, Password, NoOfDogs, NoOfCats	826	Movie database
CASTING	Castid (PK), ActorNo (FK), MovieNo (FK)	3317	Movie database (M:M intersection of ACTOR and MOVIE)
RATING / COLOURTYPE	RATINGCODE, RATINGNAME, DESCCRIPTION / COLOURCODE, COLOUR_NAME	5 / 2	Movie database look-up tables
STUDENT, SUBJECT, ENROLMENT	StuId, FirstName, Surname / SubCode, SubName / SubCode+StuId (PK), EnrolDate, FinalGrade	4 / 4 / 6	Lecture 8 Part 1: M:M expansion
PATRON, TOY, LOAN	PatronNo, ... / ToyNo, Description, YearAcquired / PatronNo+ToyNo+DateBorrowed (PK), DateReturned	3 / 4 / 8	Lecture 8 Parts 1–2: identifiers and outer joins
UNI_STUDENT, UNI_TUTOR	Name	4 / 3	Lecture 8 Part 3: UNION, INTERSECT, MINUS
CUSTOMER_W8	CustNumb (PK), Name, Gender (NULL allowed), Balance, CredLim	10	Lecture 8 Part 3: literals and UNION

Week 8 quick reference

Topic	Key points
M:M expansion	Replace the M:M by a weak (intersection) entity and two M:1 relationships; the MANY end attaches to the weak entity and the relationship names are underlined. The weak entity borrows the identifiers of both parents.
Testing identifiers	Test the schema with sample data. If the same pair can repeat (same toy borrowed on different dates) add the extra attribute (DateBorrowed). A primary key must never contain NULL, so DateReturned cannot be part of it.
Oracle dates	INSERT ... VALUES (1, '21-MAY-1996') or TO_DATE('21/05/1996','dd/mm/yyyy'). Compare dates with TO_DATE too.
Subqueries	A complete SELECT in brackets inside another statement, evaluated first and once. Allowed in WHERE and HAVING. Aggregate functions are NOT allowed directly in WHERE. = needs one value, IN/NOT IN accept many.
Joins	INNER JOIN = rows with a match in both tables. LEFT OUTER JOIN = all left rows plus matches. RIGHT OUTER JOIN = all right rows plus matches. Unmatched rows show NULL; find them with IS NULL. COUNT(col of right table) gives 0 for no match.
Set operators	UNION (no duplicates), UNION ALL (keeps duplicates), INTERSECT (rows in both), MINUS (rows of the first query not in the second). Same number of columns and matching data types (TO_CHAR/TO_NUMBER to convert).

Topic	Key points
Literals	A hard-coded value in the SELECT list, e.g. SELECT name, 'Unknown' FROM customer WHERE gender IS NULL, often combined with UNION.
Constraints	UNIQUE (column or table level for several columns), CHECK (fixed validation, passes when TRUE or NULL; no subqueries, no other tables, no SYSDATE). Use look-up tables instead of CHECK when the values may change.

Practice Questions

31 questions · write the SQL or answer yourself, then check the answer key.

Section A – Modelling: Many-to-many relationships, identifiers and relational schema

Q1. [Pass] A university lets each student enrol in many subjects, and each subject has many students. For every enrolment the system records the enrolment date and the final grade.
(a) Explain why the relational model cannot represent this many-to-many (M:M) relationship directly.
(b) Draw the ERD with the M:M expanded into two M:1 relationships.
(c) State the identifier of the intersection (weak) entity.
(d) Write the relational schema with PK and FK.

Hint: Lecture 8 Part 1, slides 2-7.

Q2. [Credit] Glenferrie Toy Library records every loan of a toy to a patron (DateBorrowed, DateReturned). The LOAN table below has been created. Decide which rows are REJECTED for each candidate primary key.
LOAN rows (PatronNo, ToyNo, DateBorrowed, DateReturned):
1: 53, 27, 1 Jun, 8 Jun
2: 53, 31, 1 Jun, 8 Jun
3: 53, 27, 19 Jul, 30 Jul
4: 53, 27, 2 Aug, NULL
5: 71, 27, 11 Jun, 18 Jun
6: 53, 31, 9 Oct, NULL
(a) PK = (PatronNo, ToyNo)
(b) PK = (PatronNo, ToyNo, DateBorrowed)
(c) PK = (PatronNo, ToyNo, DateBorrowed, DateReturned)
Which one is the correct identifier and why?

Hint: Lecture 8 Part 2, slides 3-8.

Q3. [Credit] Hawthorn Fitness has members (MemberId, Name) and group classes (ClassId, ClassName, Weekday). A member can attend the same class on many different dates, and the system records how many minutes the member stayed. (a) Draw the ERD, (b) write the relational schema, and (c) create 4 sample rows of the weak entity that prove the identifier is correct.

Hint: Follow the 9-step ERD method from Lecture 8 Part 1, slide 13.

Section B – Oracle date handling

Q4. [Pass] In Oracle, create a table EVENT (eventid NUMBER primary key, eventname VARCHAR(30), eventdate DATE). Insert one row using the DD-MON-YYYY format (21-MAY-2025) and one row using TO_DATE with the format dd/mm/yyyy (14/09/2025).

*Hint: Run one statement at a time in iSQL*Plus Junior.*

Q5. [Pass] List the full name and birth date of all actors born between 1 January 1970 and 31 December 1979 (inclusive), oldest first.

Hint: Use TO_DATE so the result does not depend on the default date format.

Q6. [Pass] In the toy library data, list PatronNo, ToyNo and DateBorrowed for every loan borrowed during June 2024.

Hint: BETWEEN includes both end dates.

Section C – Subqueries

Q7. [Credit] Why does SELECT * FROM movie WHERE runtime > AVG(runtime); fail? Rewrite it so that it lists the title and runtime of every movie longer than the average, longest first.

Hint: The subquery is in brackets, runs once and returns a single value.

Q8. [Credit] Find the title and runtime of the movie(s) with the longest running time.

Q9. [Credit] List the title and release year of all movies released in the same year as 'Forrest Gump', in alphabetical order.

Hint: The subquery returns one value, so = can be used.

Q10. [Credit] List the titles of all movies that have at least one actor born in Canada. Each title must appear only once.

Hint: Subquery with IN returns many values.

Q11. [Credit] Which rating codes have an average runtime greater than the average runtime of all movies? Show the rating code and its average runtime (rounded to 1 decimal place).

Hint: Subquery inside HAVING.

Q12. [Credit] List the title and TMDB score of all movies whose TMDB score is higher than the average TMDB score of movies rated 'PG'. Show the highest score first.

Hint: The subquery has its own WHERE clause.

Q13. [Distinction] List the actor number and full name of every actor who has never been cast in any movie, using a subquery with NOT IN.

Hint: NOT IN returns no rows if the subquery contains a NULL; casting.actorno has no NULLs here.

Q14. [Distinction] Use nested subqueries to list (alphabetically) the full names of all actors who appeared in the movie with the longest runtime.

Hint: The innermost subquery is evaluated first.

Section D – Inner and outer joins

Q15. [Pass] List the full name of every actor who appeared in the movie 'Forrest Gump', in alphabetical order (join ACTOR, CASTING and MOVIE).

Q16. [Credit] Use a LEFT OUTER JOIN to list the actor number and full name of every actor who has no row in CASTING.

Hint: The unmatched rows have NULL in every column of the right table.

Q17. [Credit] Rewrite your answer to the previous question using a RIGHT OUTER JOIN (CASTING must appear first in the FROM clause). The result must be identical.

Q18. [Credit] For every movie, show its number, title and the number of cast members, including movies that have no cast (count 0). Sort by cast size (smallest first) and then title.

Hint: COUNT(*) would count 1 for a movie without cast; COUNT(column from the right table) counts 0.

Q19. [Credit] For the actors with ActorNo 3, 7, 8293 and 90633, list the full name, movie number and movie title of every movie they appeared in. Actors with no movies must still appear (title shown as NULL). Combine an INNER JOIN (casting and movie) with a RIGHT OUTER JOIN to actor.

Hint: Lecture 8 Part 2, slide 24.

Q20. [Credit] List every toy in the toy library with the surname of the patron who borrowed it and the date borrowed. Toys that have never been borrowed must also be listed. Sort by toy description, then date borrowed.

Q21. [Credit] List every student (id, first name, surname) with the number of subjects they have enrolled in, including students who have not enrolled in anything (count 0).

Q22. [Distinction] Which subjects (code and name) have no enrolments at all? Answer twice: (a) with an outer join, (b) with a subquery.

Section E – UNION, UNION ALL, INTERSECT, MINUS and literals

Q23. [Credit] Two tables hold people: UNI_STUDENT(name) and UNI_TUTOR(name). (a) List all names that appear in either table, with no duplicates. (b) List all names from both tables, including duplicates. How many rows does (b) return and which name is repeated?

Hint: UNION removes duplicates, UNION ALL keeps them.

Q24. [Credit] List the names of people who are BOTH a student and a tutor (INTERSECT).

Q25. [Credit] (a) List the students who are not tutors. (b) List the tutors who are not students. Explain why the two answers differ (MINUS).

Q26. [Credit] List all customers with their gender. When the gender is NULL display the literal 'Unknown'. Use UNION of two queries, sorted by customer number.

Hint: Table CUSTOMER_W8 is in the Week 8 Access file. (In Oracle iSQL add your own suffix, e.g. CUSTOMER_W8_5520, if you recreate it.)

Q27. [Credit] List every customer number with the rating 'Good Customer' when balance <= credit limit, otherwise 'Bad Customer'. Use UNION and a literal column named Rating.

Q28. [Distinction] Produce one combined list with three columns Kind, Name and Yr containing (i) every movie released in 1994 (Kind = 'Movie', Name = title, Yr = release year) and (ii) every actor born in 1942 (Kind = 'Actor', Name = full name, Yr = birth year). The data types of the columns must match, so convert the numbers/dates with TO_CHAR.

Hint: Lecture 8 Part 3, slide 5: column data types in a UNION must match.

Section F – Constraints: UNIQUE and CHECK

Q29. [Credit] Write the CREATE TABLE statement for EXAM with these rules: stuid NUMBER primary key; name VARCHAR(30) NOT NULL; gender VARCHAR(1) must be M, F or I; test1 must be >= 0; test2 must be between 0 and 50; test1 + test2 must never exceed 100; the combination (name, gender) must be unique.

Hint: Lecture 8 Part 3, slides 11-14.

Q30. [Credit] Using the EXAM table from the previous question, state which of these INSERT rows succeed and which fail (and why):
1: (1,'Ann','F',60,40)
2: (2,'Bob','X',50,20)
3: (3,'Cy','M',-5,10)

4: (4,'Di','F',70,60)
5: (5,'Ed','M',NULL,NULL)
6: (6,'Flo','I',55,50)
7: (7,NULL,'F',10,10)
8: (1,'Gus','M',10,10)
9: (9,'Ann','F',20,20)

Hint: A row passes a CHECK when the result is TRUE or NULL.

Q31. [Credit] For each proposed constraint say whether a CHECK constraint is suitable and explain: (a) CHECK (dayname IN ('Mon','Tue','Wed','Thu','Fri','Sat','Sun')); (b) CHECK (managername IN ('Dave','Sue','Henry')); (c) CHECK (enddate > SYSDATE); (d) CHECK (credlim >= (SELECT AVG(credlim) FROM customer)); (e) CHECK (grade IN ('Good','Excellent','Poor')).

Hint: Lecture 8 Part 3, slides 15-16.